



(+91)-9046408944
abhishekroy@rnd.iitg.ac.in
Coochbehar, West Bengal, India
Department of EEE, Indian Institute of Technology Guwahati
<https://orcid.org/0000-0002-5301-9157>
<https://www.researchgate.net/profile/Abhishek-Roy-6>
<https://www.linkedin.com/in/dr-abhishek-roy-74983018a/>



ABHISHEK ROY

Research Overview

My current research interest is sequence design and error-correcting codes for various applications in wireless communication and information theory. For construction purposes, I have used tools like Boolean functions, combinatorics, discrete mathematics, etc. Also, some basic knowledge in programming languages, like C/C++, or software tools, such as MATLAB, can be of help. The open problems in this field needs broad range of mathematical tools to solve. My objective in this line of research is to explore the problems related to communication and information theory under a mathematical framework.

Employment

Nov, 2024- Present **Institute Post-Doctoral Fellow (IPDF)**
Department of EEE · Indian Institute of Technology, Guwahati

Ph.D. Thesis

Title	CONSTRUCTION OF 1-D SEQUENCES AND 2-D ARRAYS WITH FLEXIBLE LENGTHS FOR APPLICATION IN WIRELESS COMMUNICATION
Defense	12 April, 2024
Supervisor	Dr. Sudhan Majhi, Department of Electrical Communication Engineering, IISc, Bangalore, email: smajhi@iisc.ac.in
Co-supervisor	Dr. Subhabrata Paul, Department of Mathematics, IIT Patna, email: subhabrata@iitp.ac.in

Education

- 2024 **Doctor of Philosophy**
Mathematics · Indian Institute of Technology, Patna
- 2016 **Master of Science**
Mathematics · Indian Institute of Technology, Kanpur
- 2014 **Bachelor of Science**
Mathematics · Jadavpur University, Kolkata
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Subjects Learned

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|--------|---|
| Ph. D. | Graph Theory, Combinatorics, Wireless Communication, Rings and Modules, Probability and Stochastic Process, Technical Communication. |
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| M. Sc. | Real Analysis, Complex Analysis, Ordinary Differential Equation, Partial Differential Equation, Abstract Algebra, Linear Algebra, Measure Theory on Euclidean Space, Topology, C Programming and Data Structure, Several Variable Calculus, Differential Geometry, Representation Theory of Finite Groups, Mathematical Modelling, Functional Analysis, Measure Theory on Abstract Spaces, Knot Theory, Integral Equations. |
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| B. Sc. | Differential and Integral Calculus, 2D and 3D Geometry, Vector Calculus, Numerical Methods, MATLAB, Mathematica, Theory of Equations, Statics, Dynamics, Probability, Mathematical Biology, Linear Programming and Optimization. |
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Programming Skills

1. $\text{\LaTeX}$
 2. MATLAB
 3. C/C++
 4. Mathematica
 5. Python
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Fellowships and Awards

- 2020–2023 CSIR-Senior Research Fellowship
2018–2020 CSIR-Junior Research Fellowship
2011–2016 INSPIRE-DST Scholarship for Top 1% in XII Standard
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Competitive Examinations

1. Joint CSIR-UGC NET in *Mathematical Science* (2016)
 2. GATE in *Mathematics* (2021)
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Journal Publications

- 2025 |** P. Priyanshu, A. Roy, S. Paul, and S. Majhi, “Constructions of non-binary Golay complementary pairs of new lengths”, *Cryptography and Communications*, vol. 17, pp. 809–822, doi: <https://doi.org/10.1007/s12095-025-00782-x>
- 2024 |** A. Roy, and S. Majhi, “Lower bounds on the maximum cross-correlations of 2-D quasi-complementary array sets”, *Cryptography and Communications*, vol. 16, no. 2, pp. 229–247, doi: <https://doi.org/10.1007/s12095-023-00665-z>
- 2021 |** A. Roy, P. Sarkar, and S. Majhi, “A direct construction of q -ary 2-D Z-complementary array pair based on generalized Boolean functions”, *IEEE Communications Letters*, vol. 25, no. 3, pp. 706–710, doi: <https://doi.org/10.1109/LCOMM.2020.3038726>
- 2020 |** P. Sarkar, A. Roy, and S. Majhi, “Construction of Z-complementary code sets with non-power-of-two lengths based on generalized Boolean functions”, *IEEE Communications Letters*, vol. 24, no. 8, pp. 1607–1611, doi: <https://doi.org/10.1109/LCOMM.2020.2987625>
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Conference Publications

- 2025 |** A. Roy, S. Das, and S. Majhi, “Construction of MOGCS with Flexible Parameters via Extended Boolean Functions”, In *IEEE International Symposium on Information Theory (ISIT)*, Ann Arbor, Michigan, USA, pp. 1–6, doi: <https://doi.org/10.1109/ISIT63088.2025.11195398>
- 2024 |** A. Roy, S. Majhi, and S. Paul, “Systematic Construction of Golay Complementary Sets of Arbitrary Lengths and Alphabet Sizes”, In *IEEE Information Theory Workshop (ITW)*, Shenzhen, China, pp. 430–435, doi: <https://doi.org/10.1109/ITW61385.2024.10806962>
- 2023 |** A. Roy, and S. Majhi, “Direct constructions of multiple shift complementary sets of flexible lengths”, In *IEEE International Symposium on Information Theory (ISIT)*, pp. 2440–2445, doi: <https://doi.org/10.1109/ISIT54713.2023.10206672>
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Articles in Preparation

2025 | A. Roy, "Construction of 2-D Z-Complementary Array Code Sets with Flexible Lengths for Different System Requirements", [Online] <https://arxiv.org/abs/2109.00970>

Scientific Affiliations

- AMS Introductory Member
 - IEEE Member
 - IEEE Young Professionals
 - IEEE Information Theory Society
 - IEEE Communication Society
 - IEEE Signal Processing Society
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Languages

1. Bengali (*Mother Tongue*)
 2. English (*Professional*)
 3. Hindi (*Intermediate*)
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Extra Curricular

- Playing Guitar
 - Drawing
 - Quizzing
 - Photography
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